



Gen AI with Element

Surendra Panpaliya

Generative AI

Gen-AI

PREREQUISITES

Participants should have:

Basic knowledge of **Python programming**

Familiarity with **APIs, JSON, and HTTP requests**

General understanding of **Machine Learning and NLP concepts**

PREREQUISITES

Awareness of **cloud platforms** (Azure or GCP preferred)

Prior exposure to **Jupyter Notebooks** or **VS Code**

Knowledge of **RESTful APIs**, **Docker**, and **Git**

Some experience with **LLMs** or **prompt engineering**

LAB SETUP REQUIREMENTS

Python 3.10+ installed (preferably in a virtual environment)

JupyterLab or **VS Code** with Python plugin

Access to:

Azure OpenAI API key (or)

Google GenAI credentials (Vertex AI Studio, PaLM/Gemini API)



LEARNING OUTCOMES

Explain key concepts in

Generative AI and

Transformer-based LLMs

Build and interact with

OpenAI/Gemini models using Python



LEARNING OUTCOMES

Design **RAG pipelines** integrated with

LangChain + Milvus

Apply structured prompt

engineering strategies in LLM apps



LEARNING OUTCOMES

Utilize Walmart's internal

**LLM Gateway and evaluation
platforms**

Create simple **Agentic
applications**

with planning, execution, and tools

LEARNING OUTCOMES



Troubleshoot common LLM issues



such as hallucination or bias



Build and present a functional



Excel-based report builder GenAI app

Agenda

DAY 1: GENAI FOUNDATION & ARCHITECTURE

DAY 2: WALMART GENAI ECOSYSTEM

DAY 3: APPLICATION DEVELOPMENT WITH GENAI

DAY 4: HACKATHON & DEPLOYMENT

DAY 1: GENAI FOUNDATION & ARCHITECTURE



Objective



Build conceptual clarity and



understanding of Generative AI,



Large Language Models (LLMs),



Walmart's GenAI technology stack.

Agenda



1. What is Generative AI?



2. Neural Generative Modeling



3. Large Language Models (LLMs)



4. Transformer Architecture & Attention Mechanism

Agenda

5. RLHF and DPO

6. Python Libraries
Overview

7. Hands-On Session

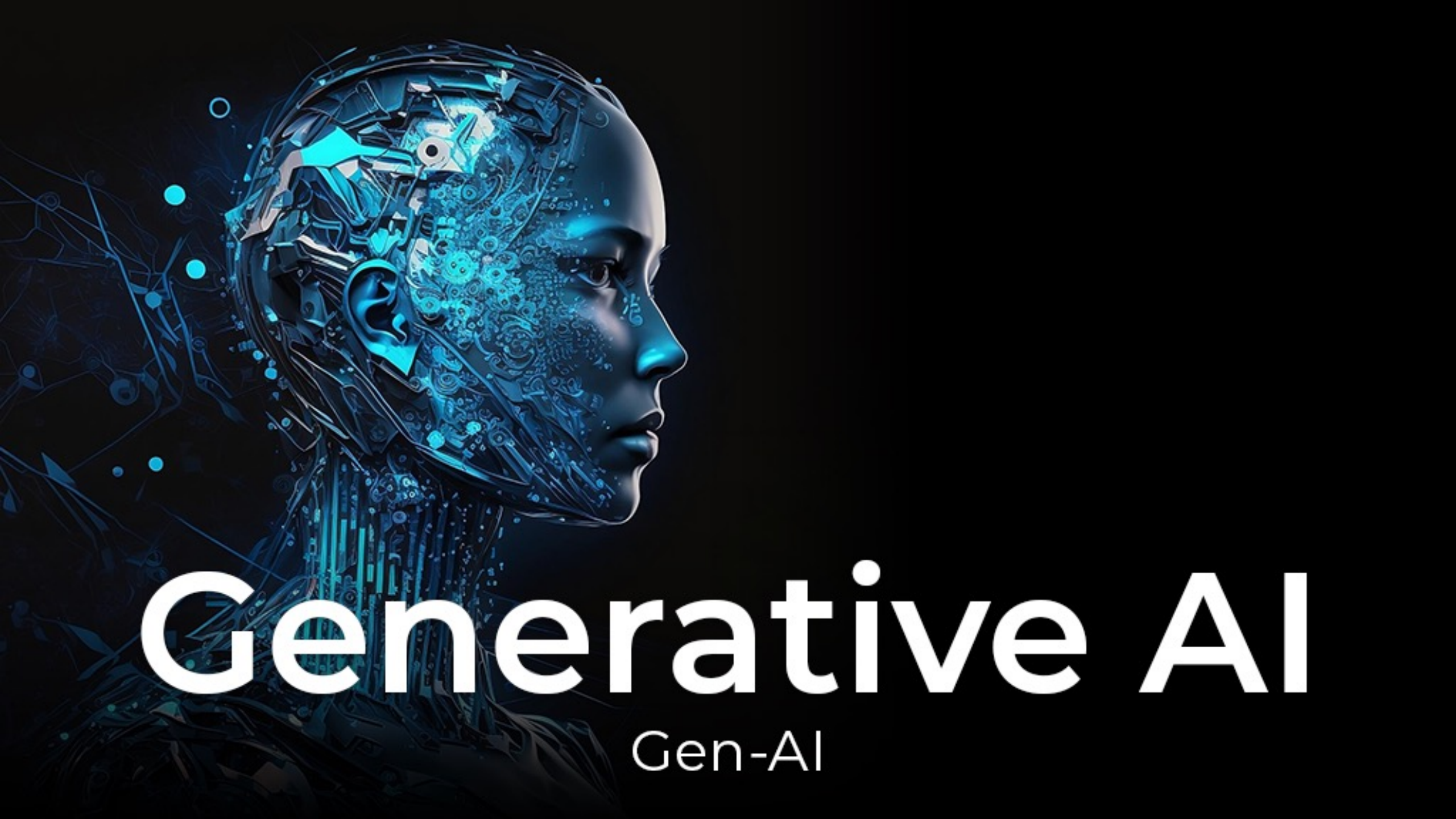
What is Generative AI?

Definition and significance in the AI landscape

Evolution from traditional AI → Deep Learning → LLMs

Demystifying the Blackbox:

How LLMs generate human-like output



Generative AI

Gen-AI

What is Generative AI?



CLASS OF ARTIFICIAL
INTELLIGENCE MODELS



CAN **GENERATE NEW**
CONTENT



SUCH AS TEXT, IMAGES,
AUDIO, VIDEO, OR CODE



BY LEARNING PATTERNS
FROM EXISTING DATA.

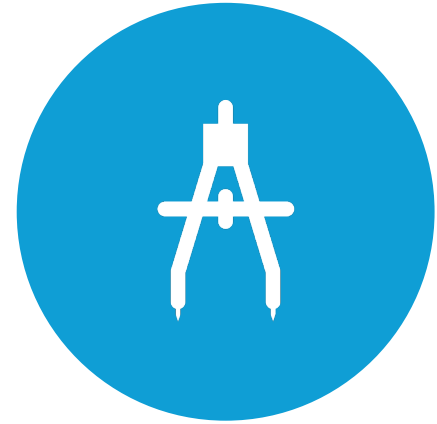
What is Generative AI?



MODELS DON'T JUST
ANALYZE DATA



THEY **CREATE**
SOMETHING NEW



MEANINGFUL FROM IT.

What is Generative AI?

At the core of GenAI are **foundation models**

GPT (for text),

DALL·E or **Stable Diffusion** (for images),

MusicLM (for audio),

Codex (for code generation), and more.

What is Generative AI?



MODELS USE DEEP
LEARNING,



ESPECIALLY
TRANSFORMERS,



TO UNDERSTAND AND
GENERATE CONTENT



THAT MIMICS HUMAN
INTELLIGENCE.

What is Generative AI?



Like teaching a very smart machine



to **read millions of examples** and



then create **something similar but original.**

What is Generative AI?



<https://www.youtube.com/watch?v=rwF-X5STYks>

Why It Matters?



**HELPS AUTOMATE
TASKS**



**THAT REQUIRE
CREATIVITY OR**



COMMUNICATION.

Why It Matters?



SAVES TIME,



REDUCES MANUAL
WORK



GIVES PERSONALIZED
EXPERIENCES.

Why It Matters?



USED IN CHATBOTS,



SEARCH ENGINES,



PRODUCT
RECOMMENDATIONS,



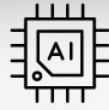
MARKETING COPY

Generative AI vs Traditional AI

Feature	Traditional AI	Generative AI
Goal	Make predictions or classifications	Generate new content: text, images, code, etc.
Input → Output	Input → Fixed output (e.g., yes/no, number)	Input → Creative output (e.g., paragraph, summary)
Training	Rule-based or statistical	Trained on vast internet-scale datasets
Example Tool	ML model to predict brake failure	LLM (e.g., GPT-4) to generate a troubleshooting guide

Evolution of AI → Deep Learning → LLMs

Stage	What It Means	Simple Analogy
Traditional AI	Rule-based systems	Like writing “if this, then that” instructions manually
Deep Learning	Learns patterns from lots of data	Like training a child to recognize cats vs dogs by showing many images
Large Language Models (LLMs)	Learns to understand and generate natural language	Like teaching an assistant to read millions of books and answer questions like a human



Artificial Intelligence



Machine Learning



Deep Learning



Generative AI

Artificial Intelligence

Broad concept of machines

doing tasks that typically require

human intelligence like

understanding language,

recognizing images, decision making

Walmart Example



A basic AI system might decide:



“If order is delayed by 3 days,



send a compensation coupon.”



Logic = manually programmed rules

Machine Learning



ML IS A **SUBSET OF AI**
WHERE



MACHINES **LEARN FROM**
DATA



INSTEAD OF BEING
EXPLICITLY PROGRAMMED.

How it works?



FEED IN DATA



TRAIN A MODEL



MAKE PREDICTIONS
ON NEW DATA

Types of ML



Supervised

Unsupervised

Reinforcement

Supervised



LABELED DATA

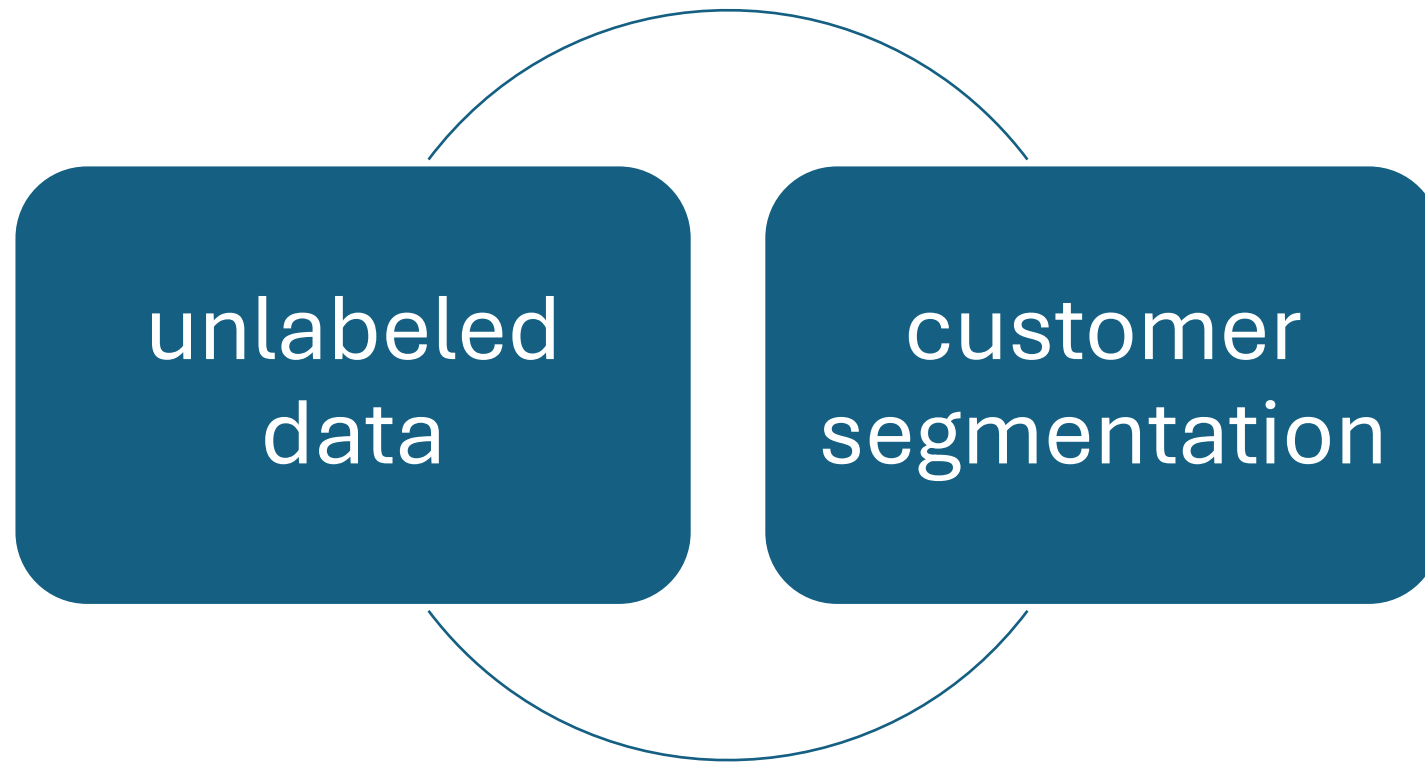


PREDICTING PRICE



BASED ON FEATURES

Unsupervised



Reinforcement



LEARNING VIA REWARDS



DYNAMIC PRICING BOTS

Walmart Example

Train a model to predict:

Will this customer return a product?

based on past behavior, product type, and location.

Learns patterns from historical data

Deep Learning (DL) – Neural Networks with Many Layers

Subset of ML that

uses **artificial neural networks**

(like the human brain)

with **multiple layers**

to solve complex problems.

Strengths



Handles huge datasets



Great for unstructured data (text, images, speech)

Walmart Example



Use DL to analyze:



Product reviews and extract sentiment:



“This shoe feels uncomfortable” → 
Negative



“This dress is amazing!” →  Positive

Generative AI

A branch of DL that focuses on

generating new content

like text, images, music, or code

by learning from existing data.

Examples

ChatGPT,

DALL·E,

Midjourney,

GitHub Copilot

Walmart Example



Automatically write product descriptions for 10,000+ items:



This microwave oven features



30L capacity, 5 cooking modes,



and child lock safety.

Large Language Models



**MASSIVE GENERATIVE
AI MODELS**



**TRAINED ON BILLIONS
OF TEXT EXAMPLES**



**TO UNDERSTAND AND
GENERATE**



**HUMAN-LIKE
LANGUAGE.**

Examples



GPT (OpenAI)



Claude
(Anthropic)



Gemini (Google)



LLaMA (Meta)

Walmart Example



LLM chatbot answers:



When will my order arrive?"



--



Your order will reach you in Pune by Tuesday.



Tracking ID: #923XX

Walmart Example



Learns context,



tone, facts



Generates clear,



natural replies

AI vs ML vs DL vs GA vs LLMs

Feature	AI	ML	DL	GA	LLMs
Definition	Machines mimicking human intelligence	Learning from data	Neural networks with layers	Generate content from data	Giant models trained on text
Input	Rules / logic	Data + Labels	Lots of data (text, images)	Text, images, music	Massive text datasets
Output	Actions / decisions	Predictions / labels	Features / patterns	New content (text, image)	Fluent, context-aware language

AI vs ML vs DL vs GA vs LLMs

Feature	AI	ML	DL	GA	LLMs
Walmart Use Case	Rule-based discount logic	Predict product return	Understand sentiment in reviews	Auto-write product copy	Answer customer queries, search
Human-Like Creativity	✗	✗	⚠ Limited	✓ Yes	✓✓ Real-time fluent responses
Example Model / Tool	Rule engine, chatbot	scikit-learn, XGBoost	CNN, RNN, BERT	DALL·E, Copilot, ChatGPT	GPT-4, Claude, Gemini, LLaMA

Imagine building a Walmart Assistant

Layer	Role
AI	“Do X if Y happens” — a fixed, rule-based helper
ML	Learns from past orders to predict behavior
DL	Sees patterns in reviews/images to detect issues
GA	Writes personalized product ads in seconds
LLM	Talks to customers like a trained human agent

Short Video

-  [AI vs ML vs DL vs Generative AI – Quick Visual Explanation](#)
- (Useful for learners and corporate teams)

How LLMs Work?

(Demystifying the Black Box)

Imagine a **super-intelligent Walmart associate** that has:

Read every Walmart product manual,

customer query, receipt, and

return policy

How LLMs Work ? (Demystifying the Black Box)



UNDERSTOOD HOW
PEOPLE ASK QUESTIONS



WHERE'S MY ORDER?



WHAT'S THE RETURN
POLICY?

How LLMs Work?

Learned to give helpful answers,

even summarizing reviews or

translating product info

into Spanish or Hindi

How LLMs Work?

That's what an LLM does.

It doesn't just **search** information,

it **understands context** and

generates human-like responses.

Walmart's GenAI Stack in Action

Scenario

A customer asks:

Is this T-shirt available in medium, and
how fast can it be delivered?

What Happens Behind the Scenes

Step	Tech Involved	Explanation
1. Query Understanding	LLM (e.g., OpenAI GPT or Walmart-trained model)	The model understands the question: size + availability + delivery
2. Information Fetching	Walmart's Product API + Inventory DB	It checks stock in nearby warehouses or stores

What Happens Behind the Scenes

Step	Tech Involved	Explanation
3. Response Generation	LLM formats a human-like answer	“Yes, this T-shirt is available in Medium and can be delivered to Pune by Tuesday if you order now.”
4. Personalization	GenAI + Customer Context	If the user is a loyal member, it might offer: “Free delivery for Walmart+ members”

Benefits to Walmart



Reduces manual queries to call centers



Enhances customer satisfaction



with **24/7 smart assistance**



Enables **multilingual support** and



auto-generated product summaries

Neural Generative Modeling



TOKENIZATION: BREAKING
DOWN INPUT TEXT



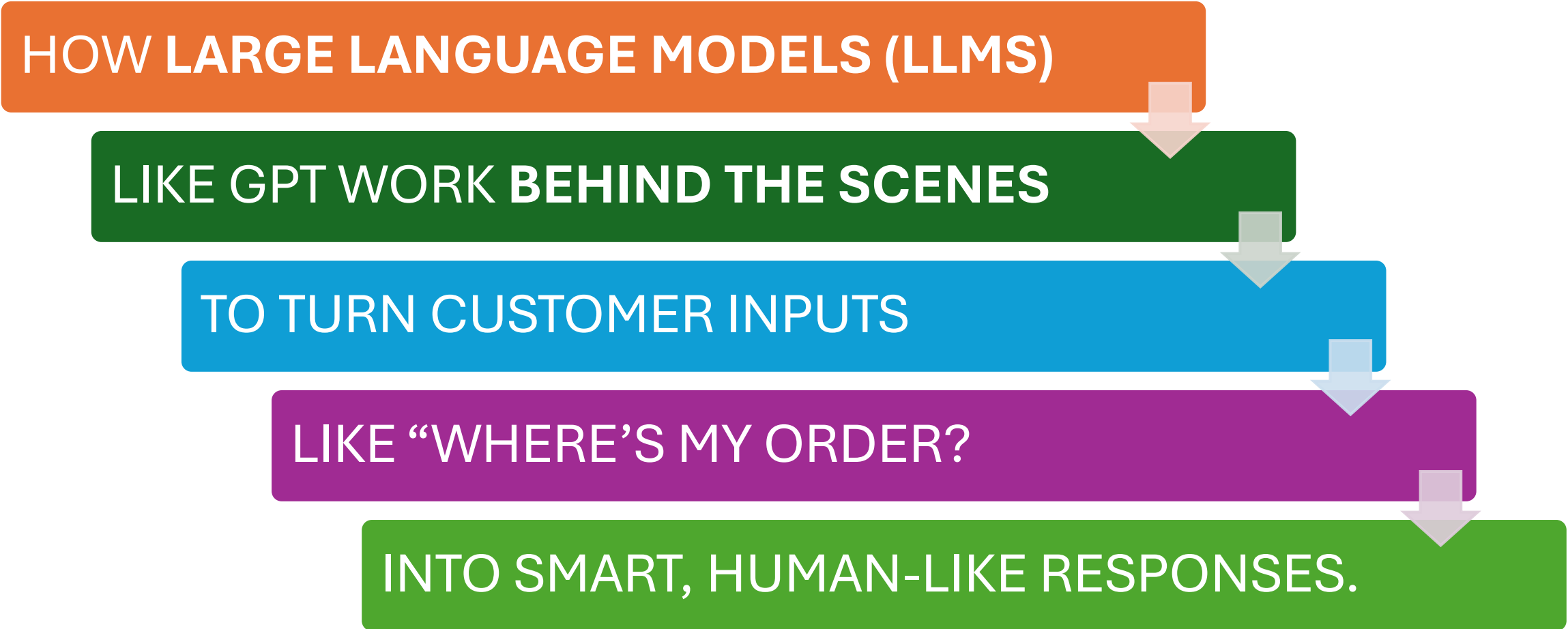
EMBEDDING: REPRESENTING
LANGUAGE IN VECTOR SPACE



GENERATION: SAMPLING AND
DECODING TECHNIQUES

What is Neural Generative Modeling?

HOW LARGE LANGUAGE MODELS (LLMS)



```
graph TD; A[HOW LARGE LANGUAGE MODELS (LLMS)] --> B[LIKE GPT WORK BEHIND THE SCENES]; B --> C[TO TURN CUSTOMER INPUTS]; C --> D[LIKE "WHERE'S MY ORDER?"]; D --> E[INTO SMART, HUMAN-LIKE RESPONSES.]
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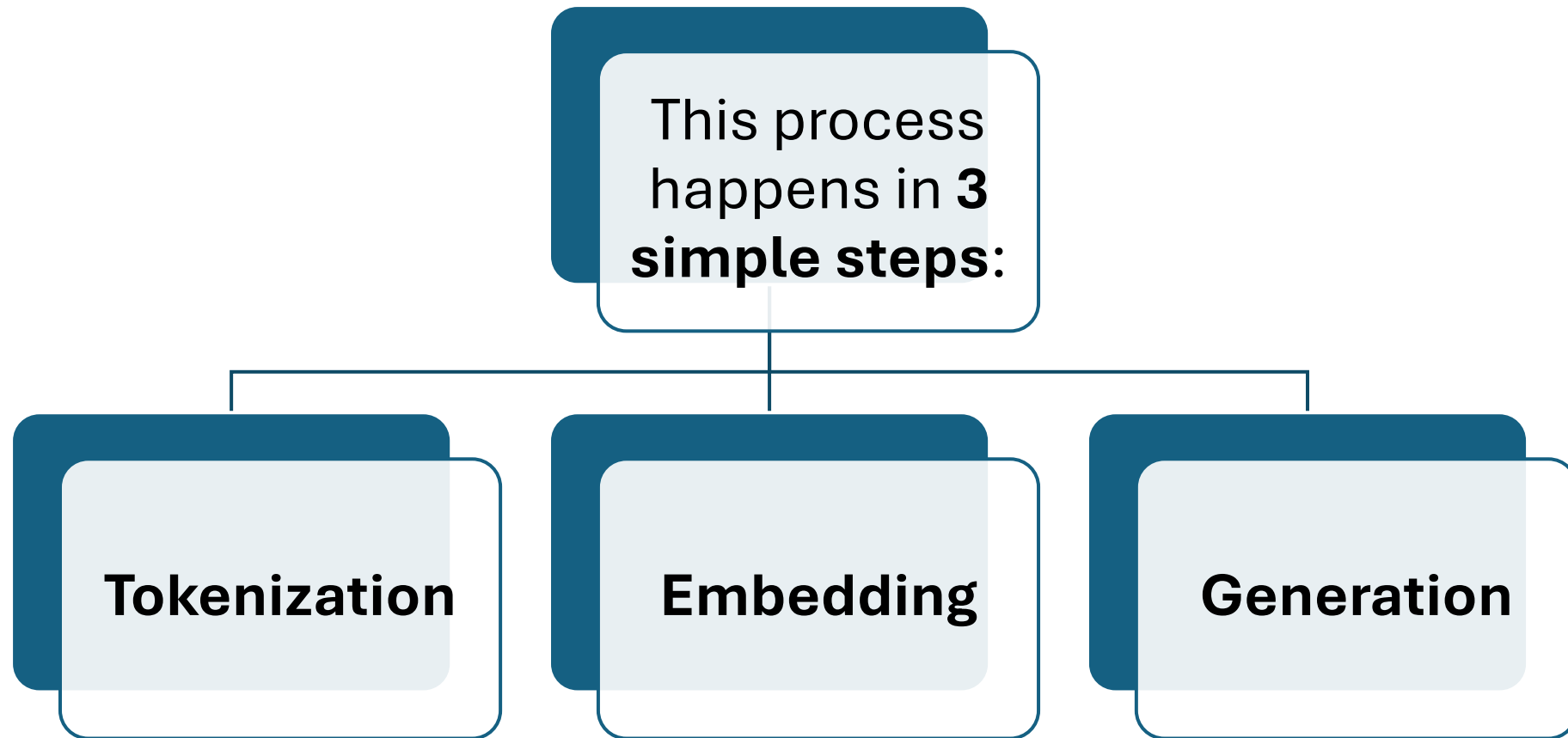
LIKE GPT WORK BEHIND THE SCENES

TO TURN CUSTOMER INPUTS

LIKE “WHERE’S MY ORDER?”

INTO SMART, HUMAN-LIKE RESPONSES.

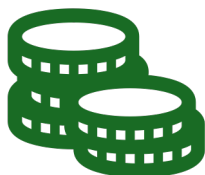
What is Neural Generative Modeling?



Step 1: Tokenization



Break the input
sentence



into **smaller**
units (tokens)



like words or
parts of words



that a computer
can understand.

Walmart Example:

A customer types:

Is this laptop available in Mumbai?

The model **tokenizes** this into:

["Is", "this", "laptop", "available", "in", "Mumbai", "?"]

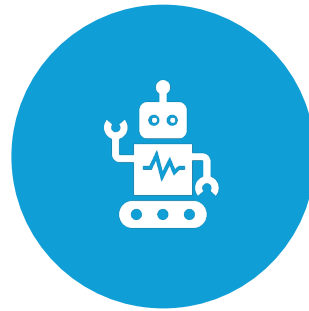
Step 2: Embedding



CONVERT EACH TOKEN
INTO A



**VECTOR (A LIST OF
NUMBERS)**



SO THE MODEL CAN
“UNDERSTAND”



ITS **MEANING AND
CONTEXT**
MATHEMATICALLY.

Step 2: Embedding



These vectors capture **relationships** between words.



For example,



“Mumbai” will be **closer in meaning**



to “city” or “location”



than “price” in this vector world.

Walmart Example

The model now knows that

“Mumbai” is a **city**, and “available”

means **check stock**, and “laptop”
is a **product**.

So it prepares to **look up inventory
and context**.

Step 3: Generation



The model generates a smart response



using **decoding techniques** like:



Greedy Search



choosing most probable word at each step

Step 3: Generation

Beam Search

considering multiple likely
responses

Top-k or Top-p sampling

for diversity

Walmart Example

Now, based on internal inventory and location info,

it generates:

“Yes, this laptop is in stock at our Mumbai store and
can be delivered by Monday.”

Real Impact for Walmart

Component	Example in Walmart GenAI Tech Stack
Tokenization	Parsing customer queries: “refund”, “late order”, “COD?”
Embedding	Understanding what “return window” or “express shipping” means
Generation	Creating replies like “Your refund will be processed in 3 days.”

Real Impact for Walmart



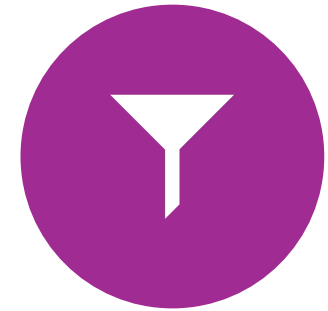
THESE MODELS ARE ALSO
USED IN:



AUTO-GENERATED
**PRODUCT
DESCRIPTIONS**



**CUSTOMER SERVICE
CHATBOTS**



SMART SEARCH (“CHEAP
SHOES FOR KIDS” →
FILTERS BY PRICE + AGE)

Summary

Step	What It Does	Walmart Use
Tokenization	Break text into pieces	“Where’s my order?” → [Where, is, my, order]
Embedding	Convert to math meaning	Understand “order” = “delivery” context
Generation	Produce response text	“Your package will arrive by 6 PM today.”

3. Large Language Models (LLMs)

Overview of popular models:

GPT (OpenAI)

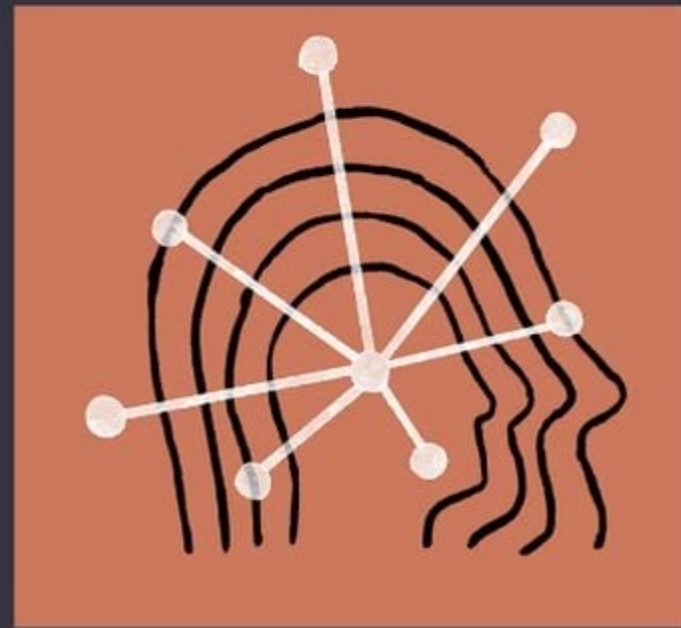
Claude (Anthropic)

Gemini (Google)

LLaMA (Meta)

The Gemini logo is displayed on a black rectangular background. It features the word "Gemini" in a sans-serif font, with the "G" in blue and the remaining letters in white. A small white four-pointed star is positioned above the "i".

Gemini



Gemini vs ChatGPT vs Claude

LLMs: GPT, Gemini, Claude



Objective:



Help decision-makers understand how to



leverage LLM like GPT (OpenAI),



Gemini (Google), and Claude (Anthropic)



to drive innovation, efficiency, and competitive edge in automotive.

What Are LLMs?



AI SYSTEMS TRAINED



ON MASSIVE
AMOUNTS OF DATA



BOOKS, MANUALS,
WEB CONTENT, CODE

What Are LLMs? (In Simple Words)



UNDERSTAND
NATURAL
LANGUAGE



ANSWER
QUESTIONS



GENERATE TEXT,
CODE,
DOCUMENTATION



PERFORM
REASONING AND
SUMMARIZATION



ACT AS A “CO-
PILOT” FOR WORK

Top 3 LLMs at a Glance

Model	Developed By	Key Strengths	Unique Features
GPT-4	OpenAI (Microsoft-backed)	Industry-leading accuracy & creativity	Plugins, Code Interpreter, Copilot integration
Gemini	Google (DeepMind)	Multimodal (text + image + video)	Deep GCP integration, powerful search context
Claude	Anthropic	Safer, ethical, large context window (100k+)	Friendly tone, long document understanding

The Battle of AI Models: GPT-4 vs Gemini vs Claude vs LLaMA

GPT-4

Gemini

Claude

LLaMA

GPT vs. Gemini vs. Claude vs Llama

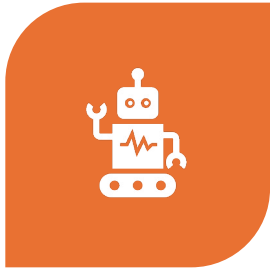


Which AI is better in 2025?



<https://www.youtube.com/watch?v=5KiDabAa9JY>

GPT-4 (OpenAI): The Powerhouse



MOST ADVANCED
LARGE LANGUAGE
MODEL,



KNOWN FOR
NUANCED
UNDERSTANDING,



CODE GENERATION



CONVERSATIONAL
ABILITIES.

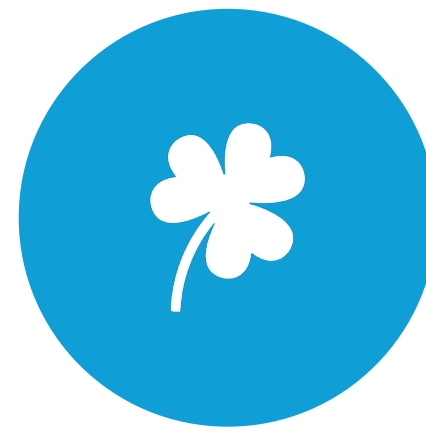
GPT-4o (“omni”)



OPENAI ADDED VOICE



VISION CAPABILITIES,



BRIDGING MULTIMODAL
EXPERIENCES.

1. Core Strengths & Use Cases

ChatGPT-4 (GPT-4o / 4.5)

Best overall performer in coding,

creative writing, and

multimodal interactions.

Excelled in calculus

problem-solving (94.7% success).

<https://openai.com/index/introducing-gpt-4-5/>

1. Core Strengths & Use Cases

Google Gemini (2.5 Pro/Flash)

Strong in multimodal context

(text+image+audio/video),

reasoning-heavy tasks, and

cost-effective free usage.

<https://gemini.google.com/app>

1. Core Strengths & Use Cases

Anthropic Claude (4 Sonnet / Opus)

Tops safety, ethical alignment,

long-context reasoning

best accuracy in reading comprehension

without hallucinations.

<https://claude.ai/onboarding?returnTo=%2F%3F>

1. Core Strengths & Use Cases

Meta Llama 4 (Scout & Maverick)

Competitive open-source option,

excels in reasoning and

massive-context tasks,

rivaling GPT-4o in benchmarks.

Meta Llama 4 (Scout & Maverick)

Meta_UI Interface

https://www.meta.ai/?utm_source=ai_meta_site&utm_medium=web&utm_content=AI_nav&utm_campaign=06112025_moment

2. Performance Highlights



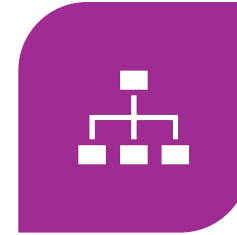
**BENCHMARKS &
ACCURACY**



**CHATGPT-4 LEADS IN
CODING & CALCULUS**



**CLAUDE PRO
FOLLOWS CLOSELY.**



**ON OLYMPICARENA,
GPT-4O AND**



**CLAUDE-3.5-SONNET
TOP GEMINI-1.5-PRO.**

3. Strengths & Weaknesses Table

Feature	ChatGPT-4/4.5	Gemini 2.5	Claude 4	Llama 4
Coding	↑ TOP Excellent	✓ Very strong	⚙ Moderate to strong	Good (via Code Llama)
Creative Writing	↑ TOP Superior prose	✓ Good (free-tier leader)	✓ Solid clarity & consistency	Moderate
Reasoning & Math	↑ TOP Top performance	✓ Strong	✓ Excellent accuracy	✓ Competitive

3. Strengths & Weaknesses Table

Feature	ChatGPT-4/4.5	Gemini 2.5	Claude 4	Llama 4
Multimodality	✓ Text + Images	↑ TOP Natively multimodal	✓ Image + document support	✓ Multimodal support
Safety & Ethics	✓ Good	! Moderate	↑ TOP Rigorous guardrails	Open-source caution

3. Strengths & Weaknesses Table

Feature	ChatGPT-4/4.5	Gemini 2.5	Claude 4	Llama 4
Cost & Access	Paid tiers	Free-tier strong	Free/open beta	Open-source
Long Context Handling	✓ Large	✓ Million-token window	✓ Opus with 200k+	✓ 1M token window

4. Emerging Trends & Developments



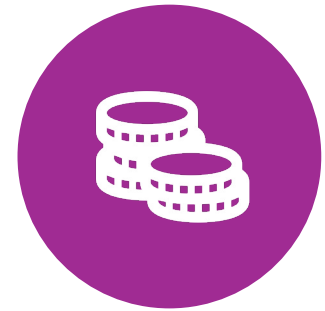
GEMINI 2.5 PRO



INTRODUCES
“DEEP THINK”



REASONING
MODE AND



MILLION-TOKEN
CONTEXTS.

4. Emerging Trends & Developments



**CLAUDE 4
SONNET/OPUS**



ADDS CODE
EXECUTION AND



BROWSING WITH HIGH
SAFETY-FOCUS.

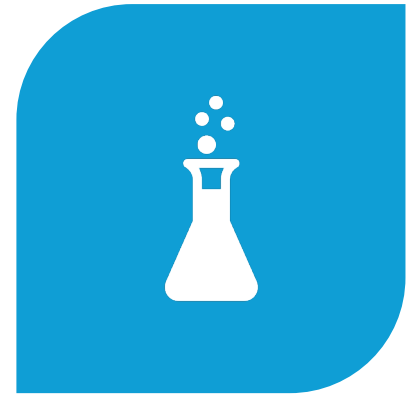
4. Emerging Trends & Developments



LLAMA 4 SCOUT/MAVERICK



ACHIEVE GPT-4 PARITY ON
REASONING BENCHMARKS



USING MIXTURE-OF-
EXPERTS ARCHITECTURE.

2025 Pick by Use Case



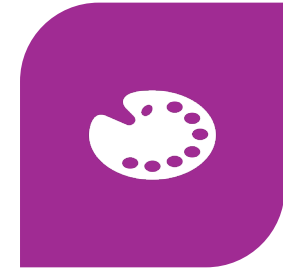
**OVERALL BEST:
CHATGPT-4,**



**WITH
UNMATCHED**



**VERSATILITY
AND**



**CREATIVE
CAPABILITY.**

2025 Pick by Use Case



**BEST FREE AND
MULTIMODAL TOOL**



GEMINI 2.5 FLASH



**STRONG
MULTIMODAL AND**



**GENERAL
PERFORMANCE.**

2025 Pick by Use Case



SAFEST AND MOST
ACCURATE

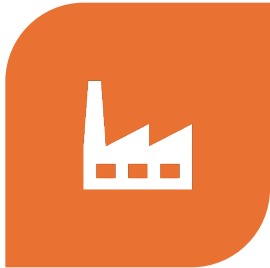


FOR LONG-FORM OR
LEGAL WORK:



CLAUDE 4.

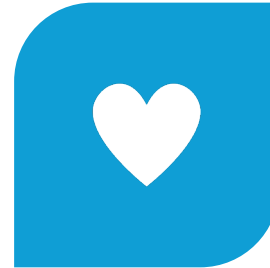
2025 Pick by Use Case



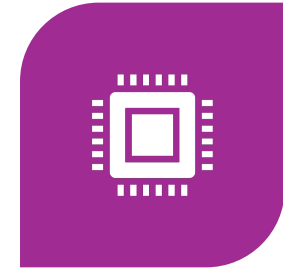
**BEST OPEN-SOURCE
POWERHOUSE**



LLAMA 4 (MAVERICK),

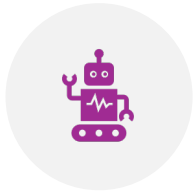


ESPECIALLY FOR



**ON-PREM/SELF-
HOSTED USE.**

Final Take



Each AI is best-in-class for specific goals:



◆ **ChatGPT-4** – elite all-rounder.



■ **Gemini** – multimodal & free utility.



■ **Claude** – safe, ethical, long-context.



■ **Llama** – scalable open-source.



How Walmart Might Use These LLMs

Model	Use Case at Walmart
GPT-4.5	Generate SEO-friendly product pages and email copy
Claude Opus	Explain complex policies or troubleshoot customer issues
Gemini Flash	Enhance store experience via voice/image queries in apps
LLaMA 4	Build proprietary, open-source solutions for internal bots

Model	Developer	Key Features	Strengths	Weaknesses
ChatGPT	OpenAI	Conversational, versatile	Strong conversational abilities, creative writing, code generation	Potential misuse, limited real-time information access
Google Gemini	Google	Multimodal, real-time information access	Technical and scientific knowledge, internet search, integrates with Google services	Can be verbose, prone to "hallucinations"
Meta AI	Meta	Contextual understanding, social interactions	Nuance in communication, social trend analysis, integrates with Meta platforms	Limited availability, potential privacy concerns
Claude	Anthropic	Reasoning, complex task handling	Analytical abilities, ethical focus, handles long-form content well	Can be slow, struggles with creative tasks

Walmart's GenAI Use Cases

GenAI Use Case	Example
 Customer Support Automation	“Where is my order?”, “How do I return this product?”
 Smart Product Descriptions & Reviews	Generate product titles, bullets, summaries
 Search & Recommendation Enhancement	“Show me kids shoes under ₹1000 in Pune”
 Supply Chain, Inventory, Pricing Insights	Forecast stockouts, restocking tips, price trends

Walmart's GenAI Use Cases

GenAI Use Case	Example
 Multilingual Chat & Accessibility Support	Translate FAQs, handle Hindi, Spanish, etc.
 Personalized Marketing & Campaigns	Tailored email, SMS copy based on user preferences
 In-store AI Assistant (voice/image/text)	“Scan product, ask price/stock via app or kiosk”

LLM Comparison for Walmart's Needs

Feature / Model	GPT (OpenAI)	Claude (Anthropic)	Gemini (Google)	LLaMA (Meta)
Model Strength	Creative writing, reasoning, coding	Safe, ethical reasoning, summarization	Multimodal (text+image+voice) + fast recall	Lightweight, open-source, fine-tuned easily
Best Use in Walmart	Chatbots, marketing, product generation	Support FAQs, policy reasoning	Mobile app assistant, search UX	Internal AI agents, warehouse automation
Accuracy (Reasoning)	✓✓✓✓	✓✓✓✓✓	✓✓✓	✓✓ (depends on fine-tuning)

LLM Comparison for Walmart's Needs

Feature / Model	GPT (OpenAI)	Claude (Anthropic)	Gemini (Google)	LLaMA (Meta)
Response Tone	Friendly + versatile	Professional + safety-first	Neutral, Google-style	Customizable
Multilingual Support	50+ languages	Limited	100+ languages + voice synthesis	Depends on training data
API Access	✅ (OpenAI platform, Azure OpenAI)	✅ (Claude API, Amazon Bedrock)	✅ (Gemini API, Google Cloud Vertex AI)	✅ (open-source via Hugging Face)

LLM Comparison for Walmart's Needs

Feature / Model	GPT (OpenAI)	Claude (Anthropic)	Gemini (Google)	LLaMA (Meta)
Multimodal (Image, Text, Voice)	GPT-4V supports images	✗ (text only)	✓ (images, speech, documents)	✗ (text only by default)
Customization	✓ (via fine-tuning, instructions)	Limited customization	Limited, focused on safety	✓✓ (open source, retrainable)
On-Premise Possibility	✗ (cloud only)	✗	✗	✓✓ (host internally)

LLM Comparison for Walmart's Needs

Feature / Model	GPT (OpenAI)	Claude (Anthropic)	Gemini (Google)	LLaMA (Meta)
Cost (for scale)	₹₹₹₹ (high, pay-per-token)	₹₹₹ (moderate via Bedrock/Claude.ai)	₹₹₹ (competitive via Google Cloud)	₹ (free, but infra cost applies)
Security & Privacy	Enterprise-grade, SOC2, GDPR	Focused on safety & data privacy	Google's enterprise-grade security	Customizable per deployment

Walmart-Specific Recommendations

Use Case	Best Model(s)	Why
Customer Chatbot / Support Agent	Claude / GPT	Claude for safe & accurate answers; GPT for flexibility
Product Content Generation	GPT	Excellent at creative text, variations, SEO content
Voice + Visual Product Search	Gemini	Multimodal capability ideal for mobile retail app UX
Warehouse AI Agent (internal use)	LLaMA	Open-source, can be tuned on logistics data

Walmart-Specific Recommendations

Use Case	Best Model(s)	Why
Policy Assistant for Managers	Claude / LLaMA	Claude for natural policy language; LLaMA for cost-effective automation
Personalized Marketing Messages	GPT	Personalization + creativity + tone tuning
Multilingual Regional FAQs	Gemini / GPT	Gemini: wider language + audio support

Access & Integration

Platform	Access URL	Integration Tips
GPT	https://platform.openai.com	Use Azure OpenAI for enterprise SLAs
Claude	https://claude.ai	Available via Anthropic API or Bedrock
Gemini	https://gemini.google.com	Use Vertex AI or Gemini Pro APIs
LLaMA	https://llama.meta.com	Download via Hugging Face for customization

Final Recommendations for Walmart



Customer-Facing:



Use GPT or Claude for **chat and content**.



Mobile/Store Assistants:



Use Gemini for **multimodal app interactions**.

Final Recommendations for Walmart



Internal/Backoffice AI:



Deploy LLaMA to save cost and
retain full control.

Final Recommendations for Walmart



For **critical reasoning + safety**,



Claude is best.



For **scale + creativity**,



GPT remains top performer.

4. Transformer Architecture & Attention Mechanism



Self-Attention and its role in LLMs



Encoder-only, decoder-only, and
encoder-decoder variations

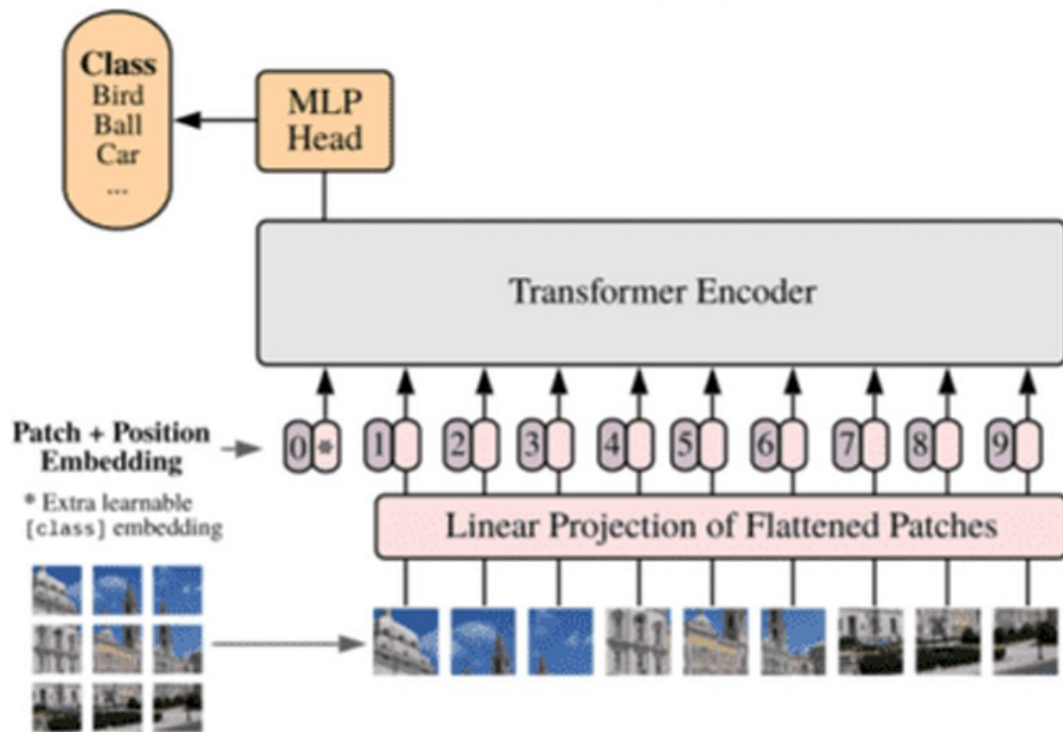


Use cases and model examples
for each type

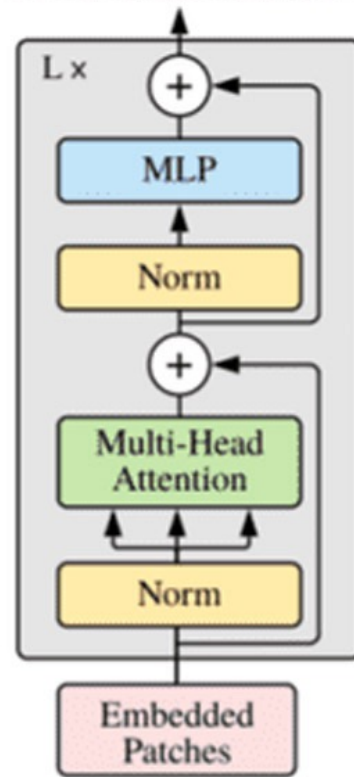


Transformers

Vision Transformer (ViT)



Transformer Encoder



What is a Transformer in LLM?



**A Deep learning
model architecture**



introduced by
Vaswani et al.



in the paper



*“Attention Is All You
Need” (2017)*

What is a Transformer in LLM?

Foundation behind LLMs

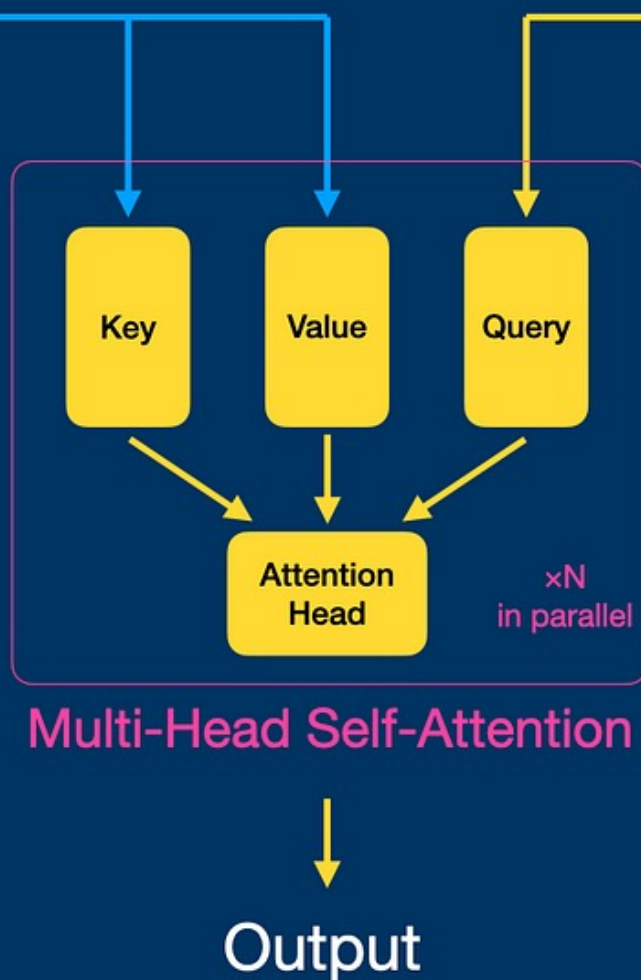
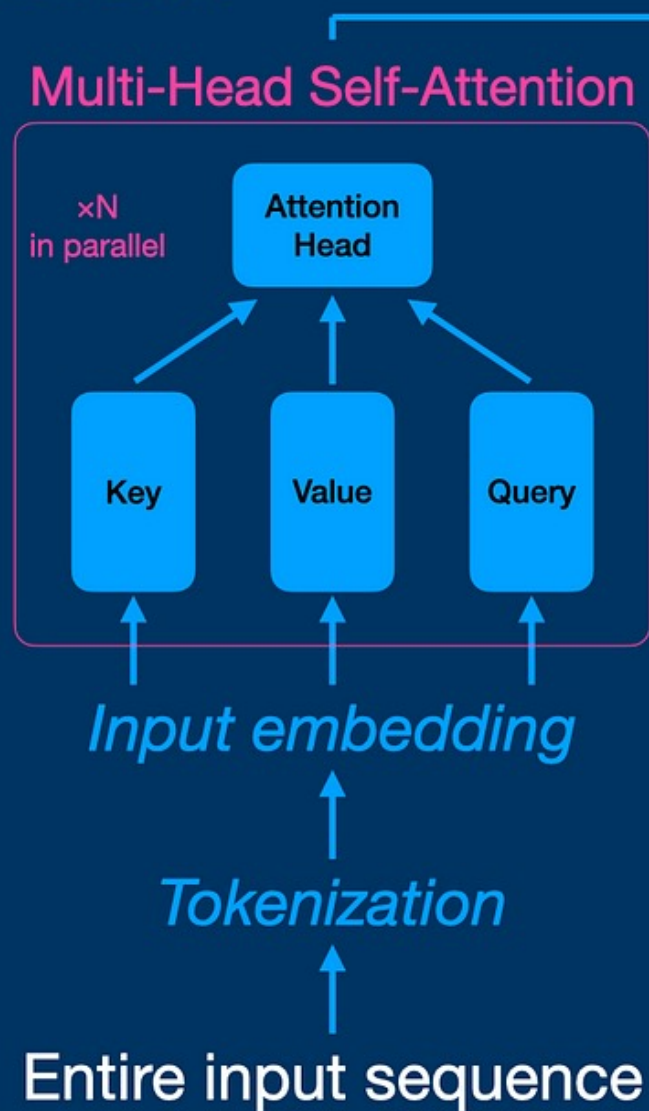
like GPT (OpenAI),

BERT (Google), and

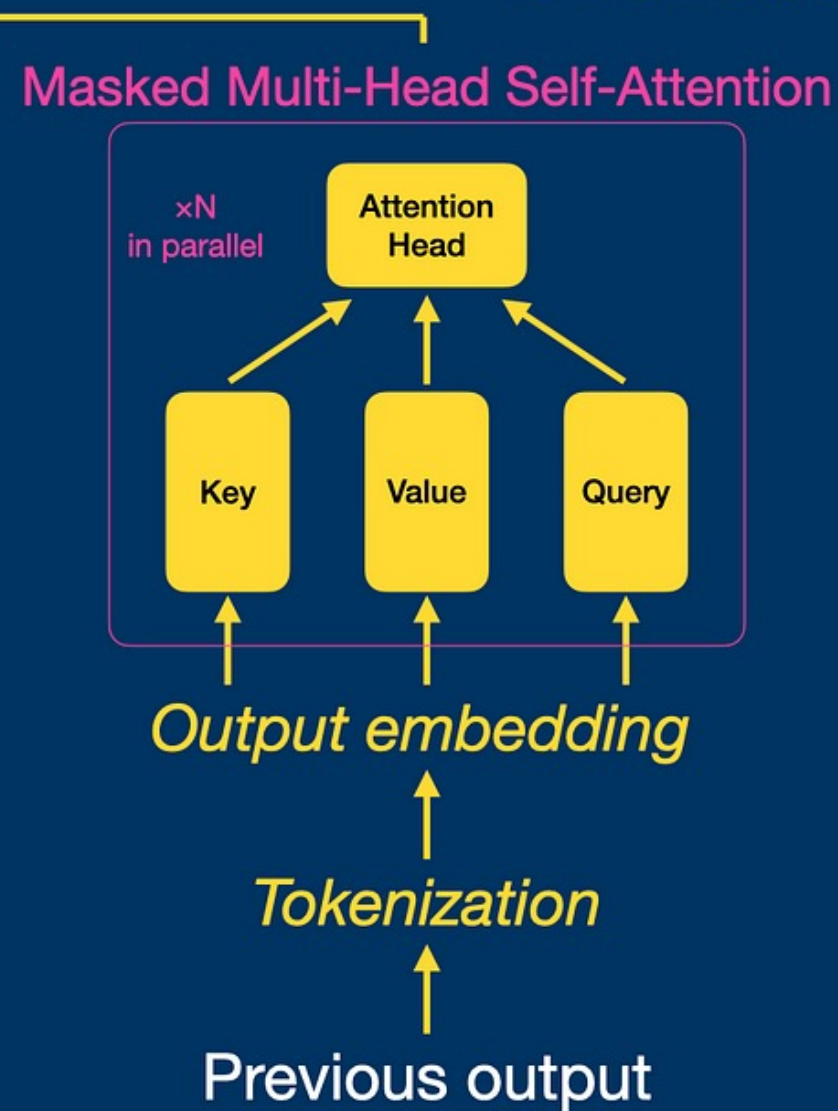
Gemini (Google DeepMind).

Main Points of Transformer Architecture

ENCODER



DECODER



*Note: many steps omitted

Self-Attention: The Core Mechanism



WHAT IT DOES?



EACH WORD EXAMINES
ALL OTHER WORDS



IN A SENTENCE TO
DECIDE WHAT MATTERS
MOST.



WORDS BECOME
**QUERIES, KEYS, AND
VALUES.**

Self-Attention: The Core Mechanism



**Why it's
important?**



Enables
understanding of
context



regardless of word
order



like correctly
relating



“it” to “laptop” or
“delivery”

Walmart Example



A question “When will my return arrive?”



“return” attends to “my” and



“arrive” to grasp intent,



rather than just the words themselves.

Transformer Variants



a) Encoder-Only



Structure: Stack of self-attention layers (like BERT)



Best for:



Understanding tasks—classification,



search embedding, document summarization.

Transformer Variants



Walmart Use:



Create vector representations



of product reviews



for sentiment analysis or

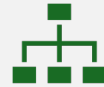


classification.

Transformer Variants



b) Decoder-Only



Structure:



Masked self-attention + feed-forward layers (like GPT)



Best for:



Generating text, chatbots, email copy.

Transformer Variants



WALMART USE:



CHAT SUPPORT,



AUTO-GENERATING
PRODUCT
DESCRIPTIONS OR



SMS NOTIFICATIONS.

Transformer Variants



c) Encoder-Decoder



Structure:



Encoder processes input →



Decoder uses both self- and cross-attention



(original “Attention is All You Need”)

Transformer Variants



Best for:



Tasks converting one text to another



translation, summarization.



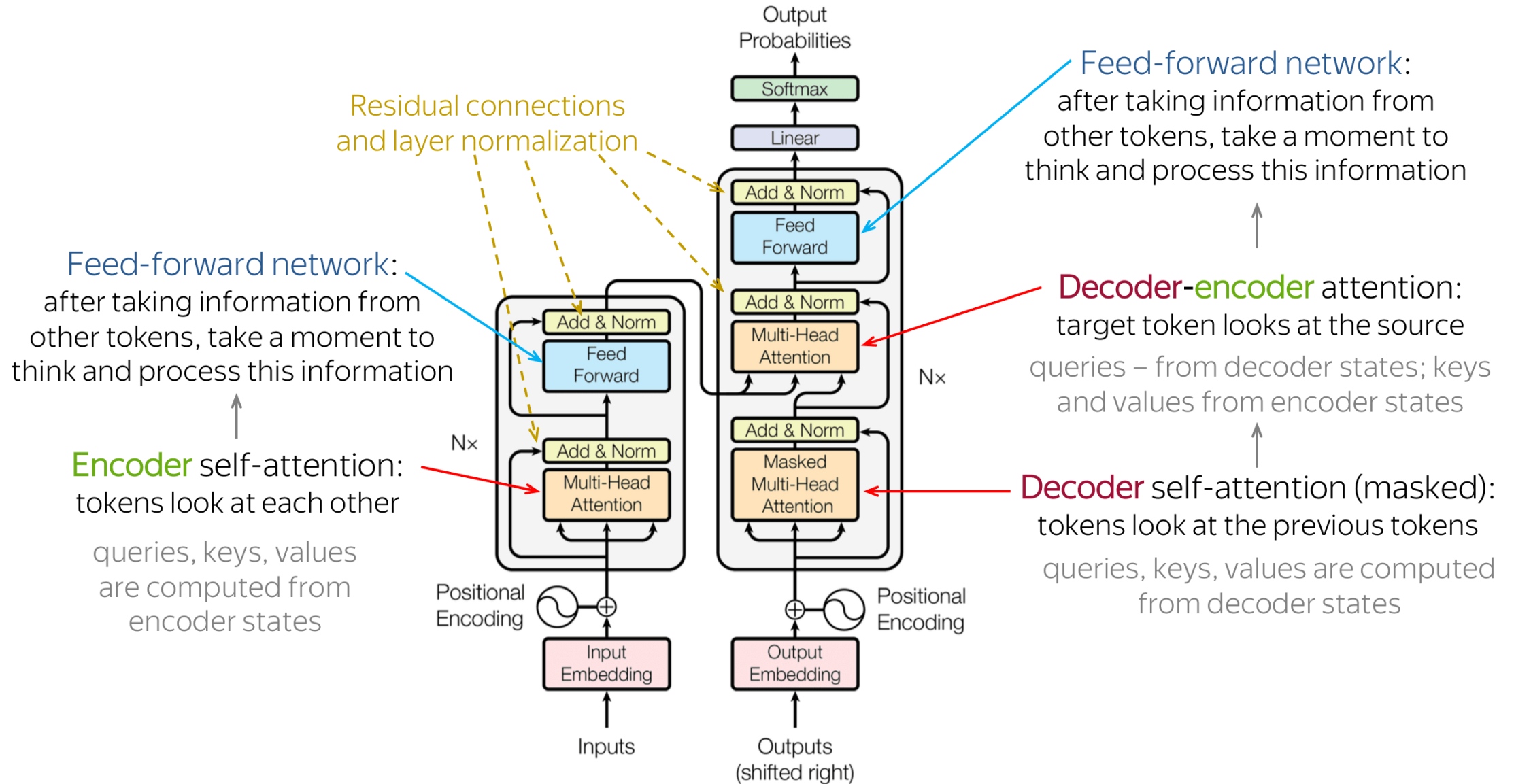
Walmart Use:



Summarize long returns policies



into a concise FAQ answer.



ENCODER (Left Side)



Takes the input sentence and



processes it into

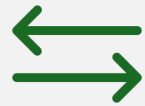


an abstract representation

Input Embedding + Positional Encoding



Words are converted into



vectors (Input Embedding).

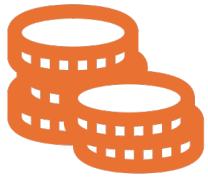


Positional information is added



so the model knows the order of words.

Encoder Self-Attention (Multi-Head Attention)



Each word (token)



**attends to all
other words**



in the sentence.



Helps understand
context.

Encoder Self-Attention Example



in “he saw a bat,”



“bat” can mean an animal or



a sports item depending



on the context.

Why Is It Called Multi-Head Attention?

Multi-head means

We split the attention into

Multiple parallel “heads”

Each head learns to focus on

different types of relationships or patterns.

Add & Norm

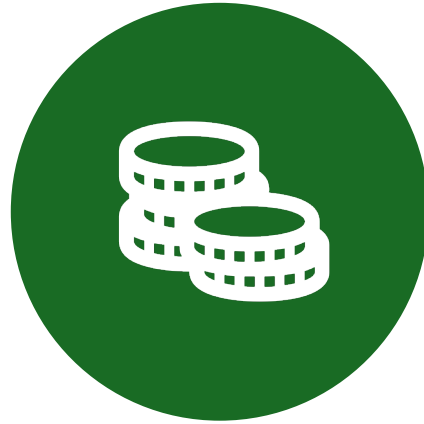
Output of attention is added back to the input (residual connection).

Layer normalization is applied to stabilize training.

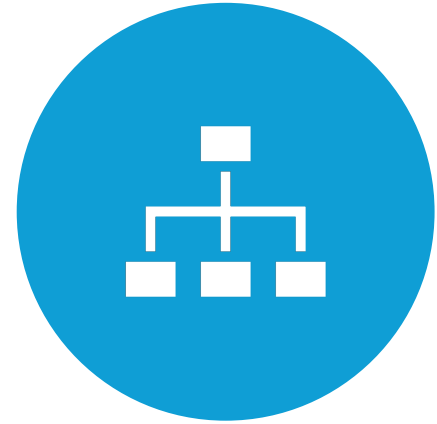
Feed-Forward Network (FFN)



A SMALL NEURAL
NETWORK APPLIED



INDEPENDENTLY TO
EACH TOKEN.

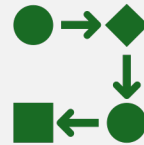


PROCESSES THE
INFORMATION FURTHER.

Add & Norm



Another residual connection and normalization.



This is **repeated N times**,



producing contextualized token representations.

DECODER (Right Side)



GENERATES THE
OUTPUT



LIKE A TRANSLATED
SENTENCE OR



NEXT TOKEN IN LLM.

Output Embedding + Positional Encoding



Previous output tokens
are embedded



shifted right for
training.



Positional encoding is
added.

Masked Multi-Head Self-Attention

Like encoder attention,

but **masked** to prevent future tokens

from being seen

Add & Norm

Decoder-Encoder Attention

The decoder token now **attends to encoder outputs**.

This is where the decoder “looks”

at the encoded input sentence

to decide what to generate.

Add & Norm

Feed-Forward Network



Further processes the information



(same as encoder's FFN).



Add & Norm



Also repeated N times.

FINAL STEP:
Output
Probabilities

A Linear layer + Softmax
converts

the final decoder output

to probabilities over
vocabulary.

FINAL STEP:
Output
Probabilities

The token with

the highest probability

is selected as

output (e.g., next word).

Encoder Flow Summary

Input → Embedding → Positional Encoding →

Self-Attention → Add & Norm →

Feed Forward → Add & Norm →

Output of Encoder Layer

Decoder Flow Summary

Output Shifted Right → Embedding →

Positional Encoding → Masked Self-Attention →

Add & Norm → Encoder-Decoder Attention →

Add & Norm → Feed Forward → Add & Norm →

Final Output → Linear + Softmax

Attention Flavors

Self-Attention:

Words attend to words

within the same sentence

(question or policy).

Attention Flavors



Masked Self-Attention:



Decoder-only models only



look at **prior words**



keeps output coherent in auto-generation

Attention Flavors



Cross-Attention:



Decoder aligns each output token



with the encoded input



from the encoder

Walmart Example



Customer asks: “Translate this review to Hindi.”



Encoder reads English review,



Decoder cross-attends to that context and



outputs Hindi translation.

Simplified Architecture Flow

Encoder-Only Flow:

Input → Tokenize →

Embed → Self-Attention →

Feed-Forward →

Output embeddings.

Simplified Architecture Flow

Decoder-Only Flow:

Prompt → Masked Self-Attention →

Feed-Forward →

Generate next word.

Simplified Architecture Flow

Encoder-Decoder Flow:

Input → Encoder → Context →

Decoder + Cross-Attention →

Output.

Walmart-Style Use Cases

Model Type	Core Feature	Walmart Example
Encoder-only	Deep understanding, embeddings	Classifying returns as ‘damaged’, ‘late’, ‘exchange’
Decoder-only	Autoregressive text generation	Creating chat responses (“Your order ships tomorrow.”)
Encoder-decoder	Translate, summarize, convert	Summarize long policy into brief customer-friendly answer

Why It Matters for Walmart?



Scalability:



Models parallelize
attention,



processing hundreds
of requests



in real-time

Why It Matters for Walmart?



Accuracy:



Capture long-distance
dependencies



“return” relates to
both



“refund” and “policy
window”.

Why It Matters for Walmart?



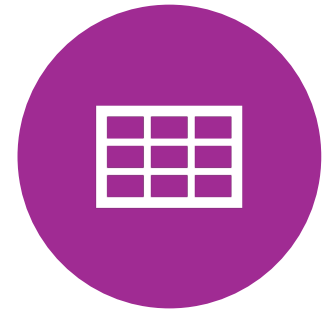
FLEXIBILITY:



APPROACH WORKS
ACROSS TASKS



CHATBOTS, SEARCH,
TRANSLATION,



SUMMARIZATION,
AND MORE.

5. RLHF and DPO



RLHF (Reinforcement Learning with Human Feedback)



Feedback loop for model alignment



DPO (Direct Preference Optimization)



Enhancing response quality without RL

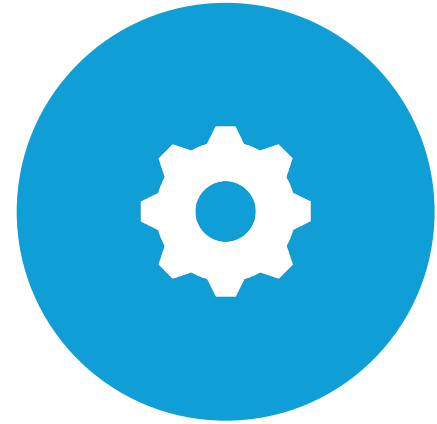
RLHF – Reinforcement Learning with Human Feedback



A PROCESS WHERE AN
LLM LEARNS



TO IMPROVE ITS
OUTPUTS BASED



ON **HUMAN
PREFERENCES.**

RLHF



Three stages:



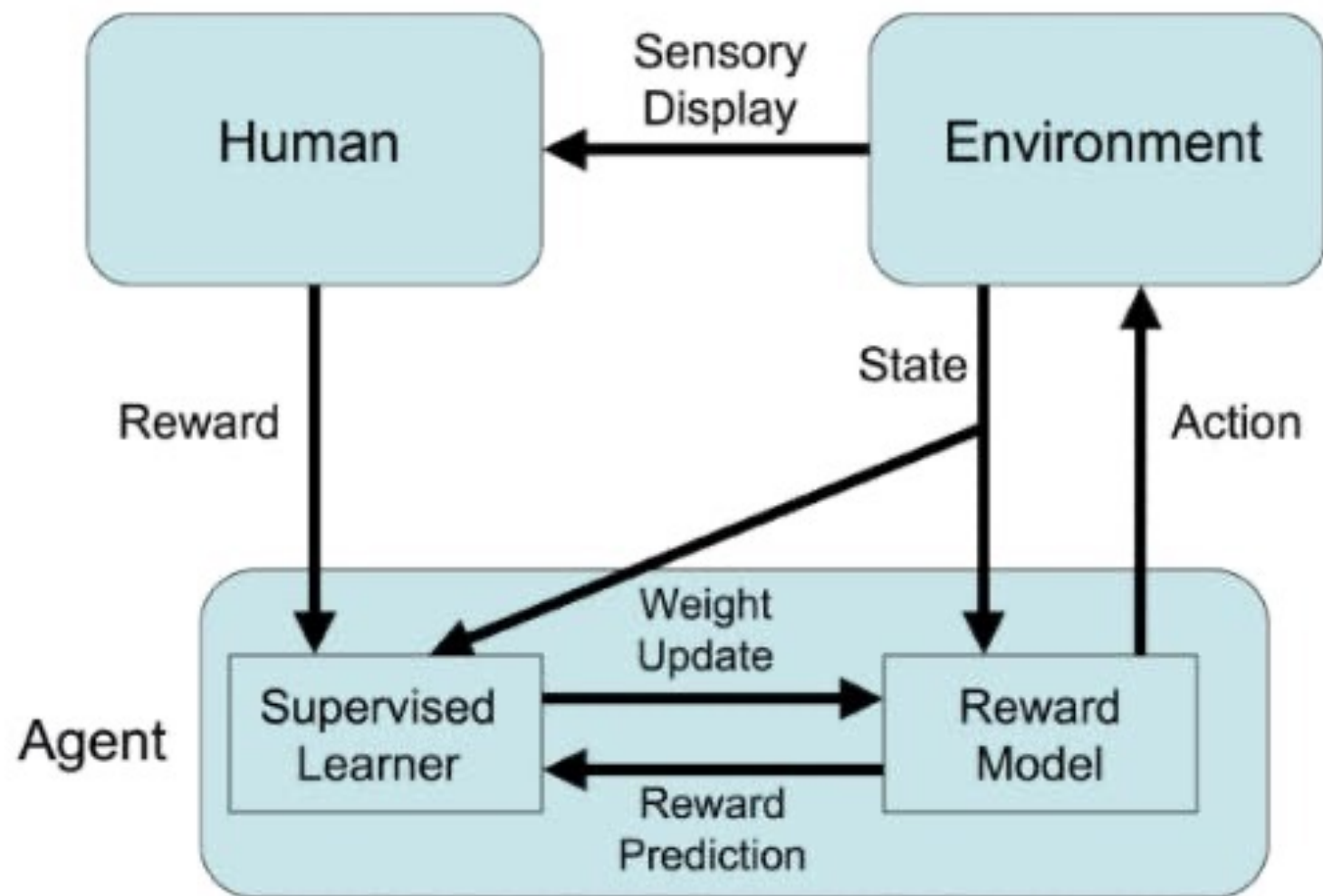
Collect examples: Humans choose better responses over worse ones.



Train a reward model to predict preferences.



Fine-tune the language model using reinforcement learning (e.g., PPO), guided by the reward model .



Example of reinforcement learning from human feedback (Image source: [cs.utexas.edu](https://www.cs.utexas.edu))

Overview of RLHF Architecture

Agent = AI model (like GPT)

Human = Provides feedback on how good
the model's actions are

Environment = Where the model acts
(e.g., conversation, task simulation)

Agent Interacts with Environment



The **agent** (AI model) performs an **action**



(e.g., generate a sentence, take a decision).



The **environment** responds by moving



to a new **state** based on that action.

2. Human Observes and Gives Feedback

The environment's state and

the agent's action are displayed

to the **human** via a **sensory
display**

(like a UI or output interface).

2. Human Observes and Gives Feedback

The human provides a reward signal

a score or ranking

indicating how “good” or “bad”

the behavior was.

2. Human Observes and Gives Feedback

Example:

A human ranks

one AI-generated sentence

as better than another.

3. Supervised Learner Gets Reward

**The supervised learner
inside**

the agent receives the

reward from the human.

3. Supervised Learner Gets Reward



It uses this feedback



to **learn preferences**



what outputs are more desirable.

Reward Model is Trained



The **reward model** takes in



the **state** (what happened) and



predicts the **reward**



that a human would give.

Reward Model is Trained



The supervised learner helps



update the reward model



using actual human feedback.

Reward Model is Trained

This lets the agent
simulate

human-like evaluations
later,

without needing a human
every time.

Agent Learns to Improve



The **reward model** starts predicting



how well the agent's actions align



with human preferences.

Agent Learns to Improve

The agent uses these
predictions

(even without human
involvement now)

to **train itself** through
reinforcement learning.

Agent Learns to Improve



The **weights** in the model are updated



to **maximize future predicted rewards.**

Loop Continues



The agent continues interacting



with the environment.



Human feedback is occasionally used



to **fine-tune** the reward model.

Loop Continues

Over time, the agent

improves its behavior

to align more closely

with human expectations.

Real-Life Example: ChatGPT

Stage	RLHF Equivalent
Environment	Chat interface
Agent	ChatGPT
Action	Generating a response
Human	A reviewer choosing best responses
Reward	Human rating or ranking
Reward Model	Predicts what kind of response a user would prefer
Supervised Learner	Fine-tunes ChatGPT using this reward signal

Walmart Example



A bot drafts two versions of a refund response:



A: “Refund processed in 3 days.”



B: “Your ₹500 refund will be credited within 3 business days to your original payment method.”

Walmart Example

Humans prefer B.

The model learns this and

gradually favors clearer,

more customer-friendly phrasing.

DPO – Direct Preference Optimization

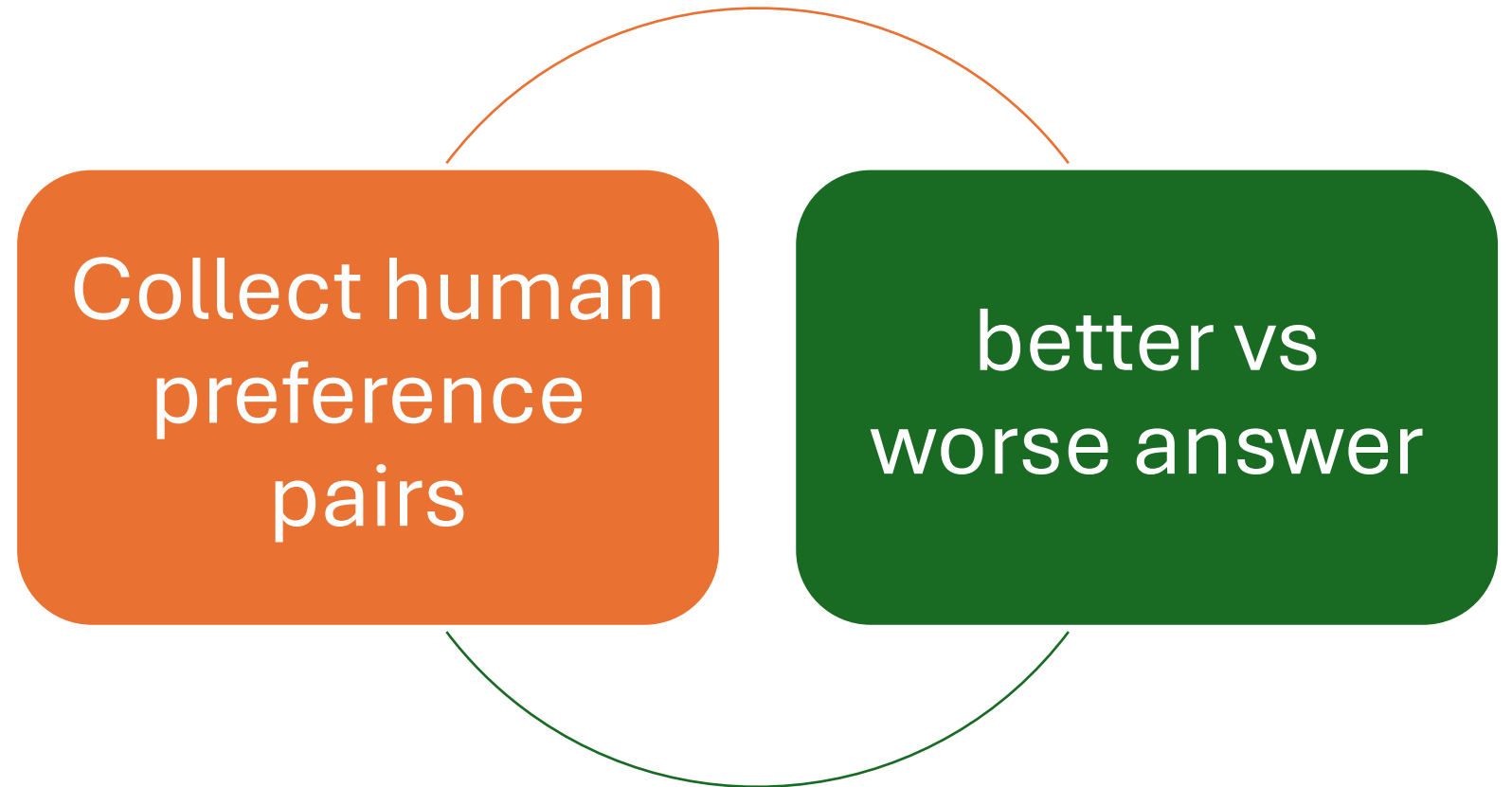
A more efficient way

to align models

with human preferences

no reward model, no RL

DPO – Direct Preference Optimization



DPO – Direct Preference Optimization

Use a **classification loss**

to directly bias the model

toward preferred outputs

Avoids RL instability and complexity.

Walmart Example



AFTER A/B RESPONSES:



USES METRIC:



$\text{LOG}(\text{PROB}(\text{A})/\text{PROB}(\text{B})).$

Walmart Example

Model is directly fine-tuned

to assign **higher probability**

to the better one

no extra RL stage needed.

Video Link



<https://www.youtube.com/watch?v=nSrj1J6ODoM>



This quick video explains



both concepts clearly and



compares their pros and cons.

RLHF vs DPO – At a Glance

Feature	RLHF	DPO
Pipeline	Complex: preference → reward → RL	Simple: direct preference → fine-tune
Computation	High (reward model + RL)	Moderate (just fine-tuning)
Stability	Can be unstable, needs tuning	More stable and easier to implement
Performance	Strong with policy flexibility	Matches or exceeds RLHF in demos

Why It Matters for Walmart?



RLHF:



Use where safety and correctness are critical



Refund policy explanations,



Sensitive customer support.

Why It Matters for Walmart?



DPO:



Ideal for scalable content tasks



like auto-generating product descriptions or



responses—with faster, leaner fine-tuning.

Summary



RLHF builds alignment via reward models +



RL—flexible but complex.



DPO directly optimizes preferences



simple, stable, efficient.

Summary

Walmart can **mix both** based on use case

RLHF for high-stakes alignment,

DPO for fast and scalable improvements.

6. Python Libraries Overview



transformers – HuggingFace for pretrained models



openai – API interface for GPT models



langchain – Orchestration for LLM workflows



vertexai – Google GenAI integration and utilities

transformers (Hugging Face)



A leading open-source library



for using and fine-tuning pretrained
transformer models

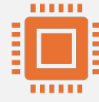


text, vision, audio



in frameworks like PyTorch, TensorFlow

Walmart Use Case



Fine-tune product-specific models



to automatically generate bullet points.



Use sentiment-analysis models



to scan customer feedback for



‘return’, ‘defective’, or ‘love’.

Why it's useful?



Thousands of ready-to-use models.



Supports training, inference,



deployment locally or in the cloud.

openai (OpenAI Python SDK)



Official library to call OpenAI models



(ChatGPT, GPT-4/4o) via API for chat,



content generation,



summarization .

Walmart Use Case

#Generate personalized marketing emails

```
from openai import OpenAI  
client = OpenAI(api_key="...")  
client.chat.completions.create(model="gpt-4",  
messages=[{"role":"user","content":"Write email about toys sale for  
Pune customers"}])
```

Why it's powerful?



Plug-and-play enterprise-grade LLMs.



Ideal for customer-facing intelligent text workflows.

langchain (LLM Orchestration Framework)



A modular toolkit to build complex applications



using LLMs—e.g., chatbots, retrieval systems,



multi-step pipelines—



with easy integration of tools like



search, databases, APIs

Walmart Use Case



Build a “Return Assistant”:



Parse user queries,



Retrieve policy from a database (RAG),



Generate an answer with GPT or Claude.

Walmart Use Case



Why it excels:



Turns simple models into powerful agents.



Supports chaining LLM calls + tool usage + memory.

vertexai (Google Cloud Vertex AI GenAI Integration)



Google's SDK to access GenAI models



like Gemini, deploy agents, and



manage multimodal workflows and pipelines .

Walmart Use Case



Deploy an in-store assistant:



Use image input (customer scans a shelf),



Recognize product via multimodal Gemini,



Respond with availability or offer details.

Walmart Use Case



Why it's useful?



Simplifies deploying secure,



scalable GenAI solutions



on Google Cloud.

Summary Table

Library	Core Role	Walmart Connection
transformers	Pretrained models local or cloud	Generate bullets, analyze sentiment, fine-tune SKU-specific models
openai	Cloud LLM access	Create chatbots, compose emails, answer customer queries
langchain	Orchestrate workflows	Build complex agents combining search, memory, LLMs
vertexai	Google GenAI deployment	Scale multimodal assistants, voice/image agents in-store or online

Why this Stack Matters for Walmart?



Speed:



Quickly deploy text generation or



question-answering with minimal code.



Scale:



Use vertexai to deploy globally with enterprise infrastructure.

Why this Stack Matters for Walmart?



Flexibility:



Compose powerful bots using langchain that tap multiple data sources.



Customization:



Fine-tune models locally with transformers to match Walmart's tone and data.

Hands-On Session



Load and interact with GPT or Gemini via Python



Tokenization → Embedding → Generation walkthrough



Explore responses with different prompt styles

Happy Learning@!!
Thanks for Your
Patience 😊

Surendra Panpaliya
GKTCS Innovations

